CSE370 Database Systems: Lab Assignment 2

Objective	Practice advanced SQL operations using a single table in MySQL.		
Due Date:	1 Week, submit before next class	**Number of Questions:**	10

Scenario:

You are a database administrator for an electronics retailer. The inventory team needs reports on pricing, discounts, suppliers, and warehouses. All operations use a single table called `products`.

Part 1: Understanding the Data

You will use a database called `Shop_<Your8DigitStudentID>`.

Example:

CREATE DATABASE Shop_12345678;
USE Shop_12345678;

The `products` table has the following attributes and data types.

Table Name: ***products***	
Attribute Name	**Attribute type**
p_id	char(4)
product_name	varchar(25)
brand	varchar(15)
category	varchar(12)
launch_date	date
price	int
discount_pct	decimal(5,3)
supplier_id	char(4)
warehouse_id	char(3)

Table data for `products` table:

p_id	product_name	brand	category	launch_date	price	discount_pct	supplier_id	warehouse_id
P001	Aero Laptop 14	TechNova	Laptop	2023-02-10	85000	0.100	S01	W01
P002	Aero Laptop 16	TechNova	Laptop	2024-05-18	120000	NULL	S01	W01
P003	Swift Notebook	Zenix	Laptop	2022-09-05	65000	0.050	S02	W01
P004	Pulse Phone X	Zenix	Phone	2024-03-21	45000	NULL	S02	W01
P005	Pulse Phone Mini	Zenix	Phone	2023-11-30	28000	0.150	S03	W02
P006	Nova Phone Pro	TechNova	Phone	2024-08-14	95000	0.200	S01	W02
P007	Buzz Earbuds	SoundWave	Audio	2022-06-12	4500	0.250	S03	W02
P008	Buzz Headphones	SoundWave	Audio	2023-01-25	12000	NULL	S03	W02
P009	Echo Speaker	SoundWave	Audio	2021-12-01	9000	0.100	S04	W03
P010	Vision Monitor 24	ViewMax	Monitor	2023-07-07	22000	NULL	S04	W03
P011	Vision Monitor 27	ViewMax	Monitor	2024-01-19	35000	0.050	S04	W03
P012	Curve Monitor 32	ViewMax	Monitor	2024-09-02	58000	0.300	S05	W03
P013	Click Mouse	TechNova	Accessory	2022-03-15	1500	NULL	S05	W04
P014	Type Keyboard	Zenix	Accessory	2023-04-04	3200	0.100	S05	W04
P015	Link USB Hub	Zenix	Accessory	2024-09-02	2400	NULL	S02	W04

Note: `discount_pct` is NULL when a product has no discount. `price` is in BDT.

Part 2: Questions

IMPORTANT: Provide one single query per question! Your query should not be hard coded, it should provide the logic mentioned in the question.

Write SQL queries to complete the following tasks: [10 * 1 = 10]

Q1. Create the products table with appropriate data types for all columns shown above.

Q2. Insert all the given data displayed in the table above into the products table.

Q3. Find the product_name, brand, launch_date, and warehouse_id of all products with the latest launch_date.

Q4. Find the product_name, p_id, price, and category of the product with the lowest price in each category.

Q5. Find the product_name, p_id, discount_pct, and warehouse_id of all products in warehouse 'W02' that have a higher discount_pct than all products in warehouse 'W01'.

Q6. Find the warehouse_id and total number of products of each warehouse that does not have any product with a price greater than 50000.

Q7. For each category, find the category, brand, and price where the price is less than at least one other brand's product in that same category.

Q8. Find the supplier_id of suppliers who do not supply any product with a price less than 5000.

Q9. Find the product_name, p_id, price, warehouse_id, and discount_pct of the product with the highest discount_pct from each supplier.

Q10. Find the category and average price of categories where the average price is less than the overall average price of all products.

Submission Instructions

Make a copy of the `submission template` in your own Google Drive.

For each task in Part 2:

1. Write the MySQL command in your submission document.
2. Execute the command in XAMPP shell (MySQL command line), take a screenshot showing the command and output, then paste it below the command in your document. **[Your StudentID with your database name, must be visible in each screenshot, ELSE YOU WILL RECEIVE 0 FOR THIS ASSIGNMENT]**
3. Repeat for all 10 tasks.
4. Download your document as PDF.
5. Rename using format: `Lab_Sec##_ID_Name_CSE370_Assignment2.pdf`

   Example: `Lab_Sec12_12345678_PeterParker_CSE370_Assignment2.pdf`
6. Submit via the Google Form here: `https://forms.gle/RfD9wtanAxuESo5L8`